DEPI - Digital Egypt Pioneers Initiative

UK Train Rides

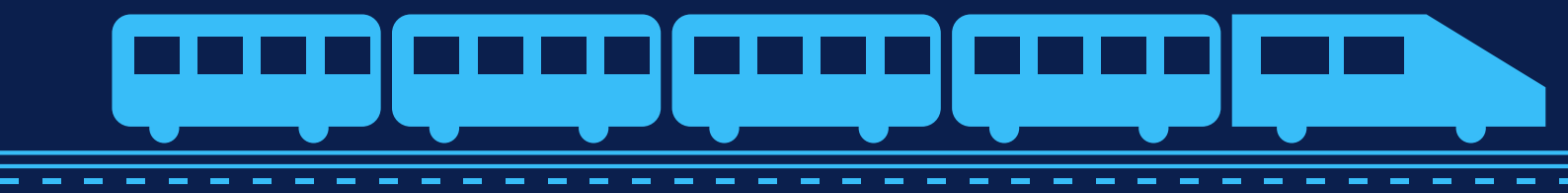
Data Analysis & Forecasting Project

Final Project Documentation - Deliverables 1 to 3

Project Planning & Management | Literature Review | Requirements Gathering

TRACK	Professional Data Analyst

TEAM MEMBERS		
Yousef Mohammed Elgabry	Adham Reda Ghazy	Mohamed Gamal Emam
Youssef Sherif Mohamed	Ahmed Nasser Abdelhafez	Mohamed Ibrahim Elgafrawy



DELIVERABLE 1 OF 3

Project Planning & Management

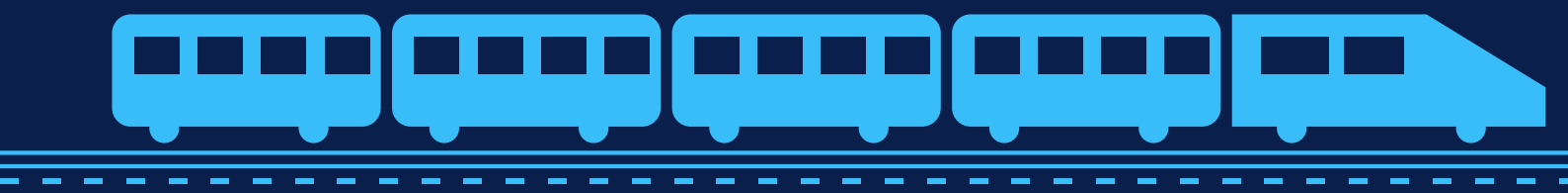
Project idea, objectives and scope

Project plan and Gantt chart

Task assignment and responsibilities

Risks and how we handle them

KPIs (success measures)



Deliverable 1: Project Planning & Management

1.1 Project Idea, Objectives and Scope

Project idea. UK Train Rides is a data analysis and forecasting project. It analyzes a dataset of UK train ticket sales and journeys to help the organization's decision makers understand ride demand, revenue, punctuality and ticket-class usage, and to forecast next month's rides, daily revenue and demand per ticket class using machine learning models. Results are delivered in a dashboard.

Objectives

- Build a data model and produce a clean, analysis-ready dataset with a documented preprocessing notebook.
- Determine all possible analysis questions that can be answered from the dataset and that matter to decision makers.
- Build and evaluate machine learning forecasting models to predict the **number of rides for the next month (May 2024)**, the **revenue for each day of that month**, and the **demand on each ticket class**.
- Build a dashboard that visualizes the answers to all analysis questions.
- Deliver a final report and presentation covering data analysis, model development and deployment.

Scope

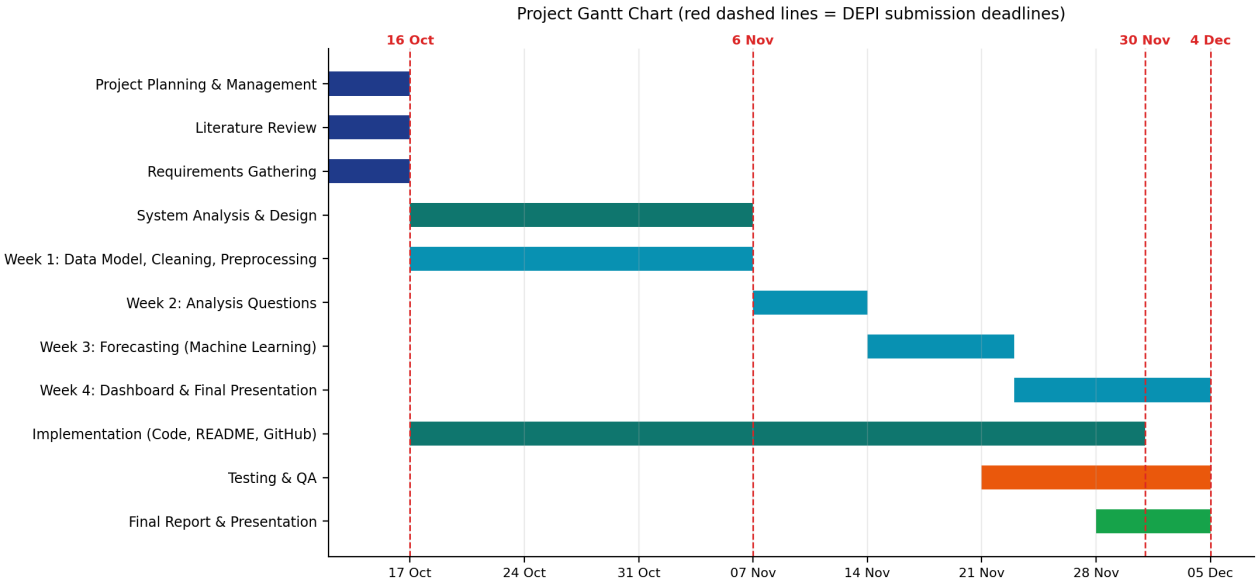
In scope	Out of scope
Data modeling, cleaning and preprocessing of the provided dataset; analysis of routes, peak times, revenue by ticket type and class, punctuality, delays, refunds and purchase behavior; forecasting of rides, daily revenue and ticket-class demand; dashboard; final report and presentation	Real-time ticketing systems; live data feeds; any data source other than the provided dataset; application development beyond the dashboard

1.2 Project Plan

Project phases (from the project brief)

Phase	Dates	Main outputs
Week 1: Data model, cleaning, preprocessing	17 Oct - 6 Nov	Cleaned dataset, preprocessing notebook
Week 2: Analysis questions	7 - 13 Nov	Set of analysis questions answerable from the dataset
Week 3: Forecasting (machine learning)	14 - 22 Nov	Forecasting models; next-month rides, daily revenue, ticket-class demand; visualization plots
Week 4: Dashboard and final presentation	23 Nov - 4 Dec	Dashboard (by 30 Nov), final report and presentation (by 4 Dec)

Project timeline (Gantt chart)



Resource allocation

Resource	Allocation
Human	6 team members (see 1.3)
Software	Python, SQL, GitHub, dashboard tool
Data	Railway ticket dataset (railway.csv) and its data dictionary
Communication	Weekly team meeting, GitHub issues and pull requests

1.3 Task Assignment & Roles

Member	Role	Responsibilities
Yousef Mohammed Elgabry	Team Leader & Dashboard	Plans the work and follows up on tasks; manages the GitHub repository and submissions; builds the dashboard with Mohamed Ibrahim Elgafrawy
Youssef Sherif Mohamed	SQL & Data Model	SQL and data model work with Mohamed Gamal Emam
Mohamed Gamal Emam	SQL & Data Model	SQL and data model work with Youssef Sherif Mohamed: tables, relationships and queries
Adham Reda Ghazy	Python	Python work with Ahmed Nasser Abdelhafez: cleaning, preprocessing and analysis in pandas
Ahmed Nasser Abdelhafez	Python	Python work with Adham Reda Ghazy: analysis and charts in pandas and Matplotlib
Mohamed Ibrahim Elgafrawy	Dashboard	Builds the dashboard with Yousef Mohammed Elgabry

Stage	Responsible
Python	Ahmed Nasser Abdelhafez, Adham Reda Ghazy
SQL / data model	Mohamed Gamal Emam, Youssef Sherif Mohamed
Dashboard	Yousef Mohammed Elgabry, Mohamed Ibrahim Elgafrawy
Forecasting (machine learning)	Whole team
Testing and documentation	Whole team

1.4 Risk Assessment & Mitigation Plan

ID	Risk	Likelihood	Impact	Mitigation
R1	Short data history: the data covers only about four months (Jan-Apr 2024), so seasonality cannot be learned and forecasts are less reliable	High	High	Rely on the weekly pattern and recent trend; test on the last weeks of data; compare with a simple baseline; state the limits of the forecast
R2	Unusual first-of-month volumes: some month-start days (e.g. 1 Jan, 1 Mar, 1 Apr) have far fewer rides than normal	Medium	High	Flag these days and test the models with and without them
R3	Inconsistent labels in the data: delay reasons spelled in different ways, and a ticket class name that differs from the data dictionary	High	Medium	Apply a documented mapping during cleaning and validate after cleaning
R4	Mock data: results describe this dataset, not real National Rail performance	High	Medium	State this clearly in the report and dashboard
R5	Missed deadlines or an unavailable team member	Medium	High	Weekly follow-up by the team leader; every stage has at least two owners
R6	Dashboard tool access or performance issues	Low	Medium	Confirm tool access early and use aggregated tables

1.5 KPIs (Success Measures)

KPI	Metric	Target
Schedule adherence	Deliverables submitted on or before the DEPI deadline	100%
Data quality	Data issues found during profiling (inconsistent labels, blanks, duplicates) resolved and documented	100% resolved; no duplicate Transaction IDs
Analysis coverage	Analysis questions answered and visualized in the dashboard	100% of defined questions
Forecast accuracy	Forecast error (MAPE) on a held-out period compared with a simple baseline	Lower error than the baseline
Forecasts delivered	Next-month rides, daily revenue for next month, demand per ticket class	All 3 delivered
Dashboard usability	Load time and clarity in a team user test	Filters respond within a few seconds
Reproducibility	Notebook and scripts run end to end from the README	Runs without errors on a clean machine

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Literature Review

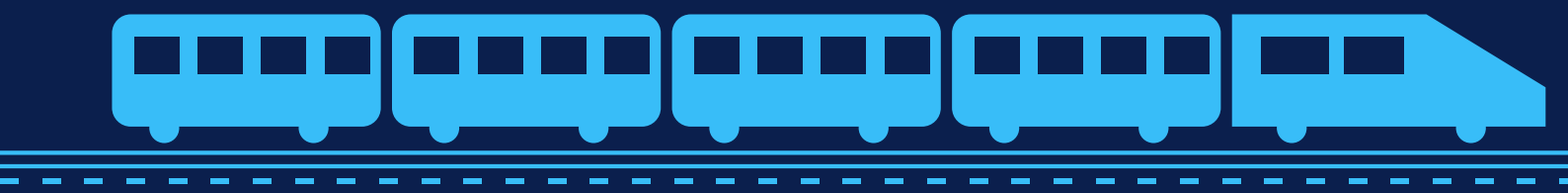
Similar projects and solutions

Pros and cons of each

Comparison with our project

Differences and what we learn

Documented sources



Deliverable 2: Literature Review

2.1 Approach

We looked for projects and solutions similar to ours, identified their strengths and weaknesses, compared them with our project, and noted how we differ and what we take from them. We searched (a) public analysis projects built on the UK Train Rides dataset in the Maven Analytics showcase, and (b) academic work on rail demand forecasting. The dashboard projects use the same dataset as ours (31,653 records, 18 fields) [1], [2]. All sources are listed in section 2.5.

2.2 Similar Projects and Solutions

A. Dashboard projects on the same dataset

Source	Tool	What it does	Strengths	Weaknesses
[3] National Rail Analysis	Power BI	Total trips, number of routes, most popular routes, revenue by ticket type and class	Answers the four challenge questions with concrete figures	Descriptive only; light data cleaning; no forecasting
[4] Insights from National Rail Data	Power BI, SQL on GitHub	Operational and financial analysis: on-time performance, revenue by class and type, with business recommendations	Turns findings into recommendations; shares SQL queries	Looks at the past only; no forecasting
[5] UK Train Data Analysis	Tableau	Top routes, railcard holders vs. non-holders, delays, peak hours, weekly average price, sales and refunds	Wide KPI set; customer segmentation by railcard; includes refunds	Descriptive only; no predictive part
[6] British Rail Performance Exploration	Excel, Tableau	Revenue, routes and punctuality exploration	Documents cleaning checks (time formats, duplicates, blanks, refund column)	Cleaning done manually in Excel, so harder to reproduce; no forecasting
[7] UK Train Rides Report	Dashboard report	Monthly analysis, one-hour time ranges, month-over-month variation, flow map of routes	Good time breakdown and route flow map	No forecasting component described

B. Academic work on rail demand forecasting

Source	What it does	Strengths	Weaknesses
[8] Fang et al. (2019), review of urban rail transit demand analysis and prediction	Categorizes demand forecasting methods: four-stage models, time series, logit regression, neural networks	Broad overview of method families	A review, not an implementation; focused on urban rail
[9] Passenger demand forecasting for railway systems	Compares neural networks and machine-learning algorithms for station-level demand using several error measures	Good model comparison with multiple error metrics	Station-level passenger data at a different scale; needs much more history than we have
[10] Boltaikhanova et al. (2024), Forecasting Railway Ticket Demand (preprint)	Forecasts railway ticket demand per origin-destination pair with ARIMA, SARIMAX and LSTM, and with a CNN that predicts all pairs together, using about 2.5 years of Kazakhstan ticket sales	Directly about ticket demand; finds a 7-day seasonality; compares models with MAE and RMSE	Needs years of data; preprint (not peer-reviewed); neural networks are heavy for a small dataset

2.3 Comparison with Our Project

Capability	Dashboard projects [3]-[7]	Academic work [8]-[10]	Our project
Descriptive analysis (routes, revenue, punctuality)	Yes	Not the focus	Yes (Week 2)
Reproducible SQL data model and documented cleaning	Partly ([4] shares SQL, [6] documents Excel checks)	-	Yes (Week 1)
Forecast of next month's rides	Not found in the reviewed projects	Yes (demand forecasting in general)	Yes (Week 3)
Forecast of daily revenue for next month	Not found in the reviewed projects	Not the main output	Yes (Week 3)
Demand per ticket class	Class revenue analyzed descriptively only	Not found	Yes (Week 3)
Interactive dashboard of all answers	Yes	No	Yes (Week 4)
Method suited to a short history	-	Assumes long history	Designed for a 121-day history, checked against baselines

2.4 Differences and What We Learn

How our project differs

- It goes beyond describing the past: it forecasts rides, daily revenue and ticket-class demand.
- It builds a SQL data model and a documented, reproducible cleaning notebook instead of manual spreadsheet edits.
- It brings descriptive analysis, forecasting and a dashboard together in one deliverable.
- It chooses forecasting methods that fit a short history rather than the long-history deep learning methods used in the academic work.

What we take from them

- From the dashboard projects: the common business questions (popular routes, peak times, revenue by ticket type and class, on-time performance), railcard segmentation, refund analysis, and linking findings to recommendations [3]-[7].
- From [6]: explicit cleaning checks for time formats, duplicates, blanks and refund amounts, which we will repeat in SQL and pandas.
- From [8]-[10]: using several error metrics and a baseline, comparing candidate methods before choosing one, and using the weekly (7-day) pattern that [10] finds in daily ticket demand.

Note: our data covers 121 days (1 Jan to 30 Apr 2024), while [10] uses about 2.5 years. We therefore use simple, robust methods and measure them against baselines.

2.5 Sources

[1] Maven Analytics. UK Train Rides dataset page. https://mavenanalytics.io/data-playground/uk-train-rides
[2] Maven Analytics. Maven Rail Challenge. https://mavenanalytics.io/challenges/maven-rail-challenge
[3] Maven Showcase project: National Rail Analysis. https://mavenanalytics.io/project/15367
[4] Maven Showcase project: Insights from National Rail Data: An Economist's Perspective. https://mavenanalytics.io/project/15458
[5] Maven Showcase project: TABLEAU Dashboard (UK Train Data Analysis). https://mavenanalytics.io/project/15487
[6] Maven Showcase project: British Rail Performance Exploration. https://mavenanalytics.io/project/15519
[7] Maven Showcase project: UK Train Rides Report. https://mavenanalytics.io/project/15610
[8] Fang, Z., Cheng, Q., Jia, R., Liu, Z. (2019). Urban rail transit demand analysis and prediction: a review of recent studies. In: Intelligent Interactive Multimedia Systems and Services (KES-IIMSS 2018), Springer, pp. 300-309.
[9] Nar, M. and Arslankaya, S. (2022). Passenger demand forecasting for railway systems. Open Chemistry, 20(1), 105-119. DOI 10.1515/chem-2022-0124. https://www.degruyterbrill.com/document/doi/10.1515/chem-2022-0124/html
[10] Boltaikhanova, T., Bostanbekov, K., Nurseitov, D., Begimbetov, O. (2024). Forecasting Railway Ticket Demand. Preprint (not peer-reviewed), Preprints.org, 1 Nov 2024. DOI 10.20944/preprints202411.0098.v1. https://www.preprints.org/manuscript/202411.0098

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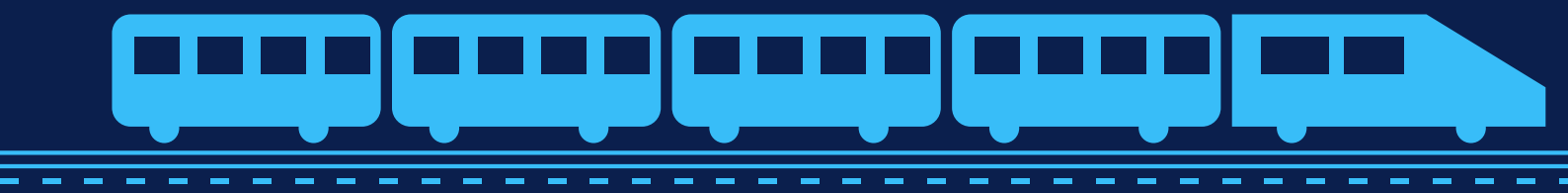
Requirements Gathering

Stakeholder analysis

User stories and use cases

Functional requirements

Non-functional requirements



Deliverable 3: Requirements Gathering

3.1 Stakeholders and Their Needs

Stakeholder	Role / interest	Needs
Organization decision makers (operations, commercial and planning managers)	Primary consumers of the analysis	Clear answers to business questions; reliable forecasts of rides, revenue and ticket-class demand
Lecturer / DEPI evaluators	Assess the project	Complete documentation, working implementation, testing, presentation, on-time submission
Project team	Design, build and deliver	Clear scope, roles, schedule and data access
Data owner / provider	Provides the dataset	Correct and responsible use of the data
End users of the dashboard	Explore the results	Easy navigation, filters, readable charts

3.2 User Stories & Use Cases

ID	User story
US1	As a decision maker, I want to see how many rides occur over time so that I can understand demand trends.
US2	As a decision maker, I want the forecasted number of rides for next month so that I can plan capacity.
US3	As a finance manager, I want the forecasted revenue for each day of next month so that I can plan revenue.
US4	As a commercial manager, I want the demand on each ticket class so that I can adjust ticket offerings.
US5	As an operations manager, I want to see delays and cancellations by cause and route so that I can reduce them.
US6	As a finance manager, I want to see refund requests linked to delays and cancellations so that I can understand their cost.
US7	As a decision maker, I want to filter the dashboard so that I can focus on the period or category I care about.

Use case	Actor	Description / outcome
UC1: View analysis dashboard	Decision maker	Opens the Tableau dashboard, applies filters and reads answers to the analysis questions
UC2: View forecasts	Decision maker	Views next-month rides, daily revenue forecast and ticket-class demand with the supporting plots
UC3: Prepare data	Data team	Loads raw data, applies cleaning and preprocessing, stores the cleaned dataset in the data model
UC4: Train and evaluate forecasting models	Data team	Builds models, validates them on held-out data and selects the best one

3.3 Functional Requirements (Functions and Features)

ID	Function / feature	Source
F1	Load the dataset into a relational data model.	Brief, Week 1
F2	Clean and preprocess the data, derive analysis fields (route, day of week, departure hour, delay in minutes, refund amount) and output a cleaned dataset with a documented, reproducible notebook. Cleaning covers: keeping "None" in Railcard as a category, standardizing delay-reason labels (8 into 5), keeping blank arrival times only for cancelled journeys, handling arrivals after midnight, and checking for duplicates.	Brief, Week 1
F3	Define the analysis questions the data can answer (routes and stations, peak travel times, revenue by ticket type and class, punctuality and delay causes, refunds, purchase behavior) and answer each with a query or computation and a visualization.	Brief, Weeks 2 and 4
F4	Forecast the number of rides for the next month (May 2024) and the revenue for each day of that month.	Brief, Week 3
F5	Forecast the demand on each ticket class (Standard and First Class).	Brief, Week 3
F6	Provide visualization plots for the forecasts and evaluate them on held-out data against simple baselines.	Brief, Week 3
F7	Provide a dashboard that visualizes the answers to all analysis questions, with filters.	Brief, Week 4
F8	Deliver a final report and presentation covering data analysis, model development and deployment.	Brief, Week 4

3.4 Non-functional Requirements (Quality)

The quality requirements are chosen to fit the nature of the project (data analysis and forecasting) and the Professional Data Analyst track.

Category	Requirement
Performance	Dashboard views respond quickly (filters within a few seconds); the full pipeline runs end to end on a standard laptop (the dataset has about 31 thousand rows).
Security	Data is used only for the project; the dataset contains no personal names (Transaction IDs only); credentials and private data are never stored in the repository.
Usability	The dashboard is clear, consistent and easy to navigate for non-technical users, with simple filters and readable charts.
Reliability and accuracy	Forecasts are validated on held-out data and compared with a baseline; data transformations are documented so results can be reproduced.
Transparency	Reports and the dashboard state that the data is mock data covering 1 Jan to 30 Apr 2024.
Track fit	The work follows the data analyst workflow (data preparation, analysis, forecasting, visualization, reporting) using the toolset in the project brief: SQL, Python (pandas, Matplotlib, scikit-learn) and a dashboard tool.